

Textbook priors over visual features

Can a vision–language model’s textbook knowledge stand in for labelled data?

Technical report, generated by the workflow

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1 The question, and what was fixed before the numbers existed

A multimodal language model carries textbook knowledge about what pathology looks like. This study asks what that knowledge is worth, measured in the currency a practitioner cares about: labelled images. On 3 MedMNIST 2D benchmarks spanning 3 modalities, a vision–language model scores each image against a cited bank of diagnostic visual features, and four arms are compared on one shared test sample:

- **A, zero-shot** — the model’s class distribution from the image and the class names. No labels.
- **B, textbook-only** — nearest class fingerprint from the concept bank. No labels.
- **C, concept probe** — regularised logistic regression on the concept scores, n labels.
- **P, pixel probe** — the same regression on frozen ImageNet features, the same n labels.

Arms C and P differ in their features and in nothing else: the same classifier, the same regularisation search and the same labelled subsets. The three hypotheses and their decision rules were fixed in `WORKFLOW.md` before any number existed, and are reported here as they were written.

2 Data and provenance

Six MedMNIST v2 224-pixel benchmarks (Zenodo record 10519652), read from the pinned release files whose names, checksums, split sizes and label maps are recorded in `config/medmnist.yaml` and held to the installed `medmnist` package by the smoke tier. Per dataset: a 500-image test sample shared by every arm and every model, and a 2000-image labelled pool from the official train split, both drawn with seed 0.

The concept bank (`data/concepts/`) was built outside the workflow by literature review, with every feature and every class fingerprint carrying a citation. **Expert review is simulated:** no clinician has reviewed those files, and each says so. Nothing downstream should be read as if one had.

A stated limit. The test sample is random rather than stratified, so a rare class is thin: dermamnist holds six dermatofibromas and five vascular lesions of its 500. Because the AUC is the unweighted mean of one-vs-rest columns, such a column counts as much as one built on hundreds, and the absolute per-dataset AUCs should be read with that in mind. The hypotheses are paired differences on the same images, where a thin column is common mode.

3 Methods as executed

Every response from the model was archived raw, with the served model name and the hash of the prompt string that produced it; everything downstream is a deterministic function of that archive. An answer that did not parse, or that named a level outside its concept’s own scale, was recorded as missing and never repaired into a value. Concept levels map to equally spaced values on $[0, 1]$; a missing answer takes the labelled pool’s median for that concept and is flagged by an indicator column. Arm B scores a class by the negative mean absolute difference over the concepts that both the fingerprint and the image commit to. Arms C and P choose their L2 strength by cross-validation inside the n labelled images, with no validation set. The metric is `medmnist.evaluator.getAUC`, and every interval is a 95% percentile interval from a paired bootstrap over the test images, shared by every arm.

4 Results

H1 — Substitution: not supported

Textbook features are worth a measurable number of labelled images. Supported if the median n_B is at least 100 **and** arm C beats arm P at the smallest labelled subset on every dataset. Arm C won on 1 of 3 datasets (one-sided sign test $p = 0.8750$); the pixel probe never reached the textbook arm inside the curve on 0 datasets and started above it on 2.

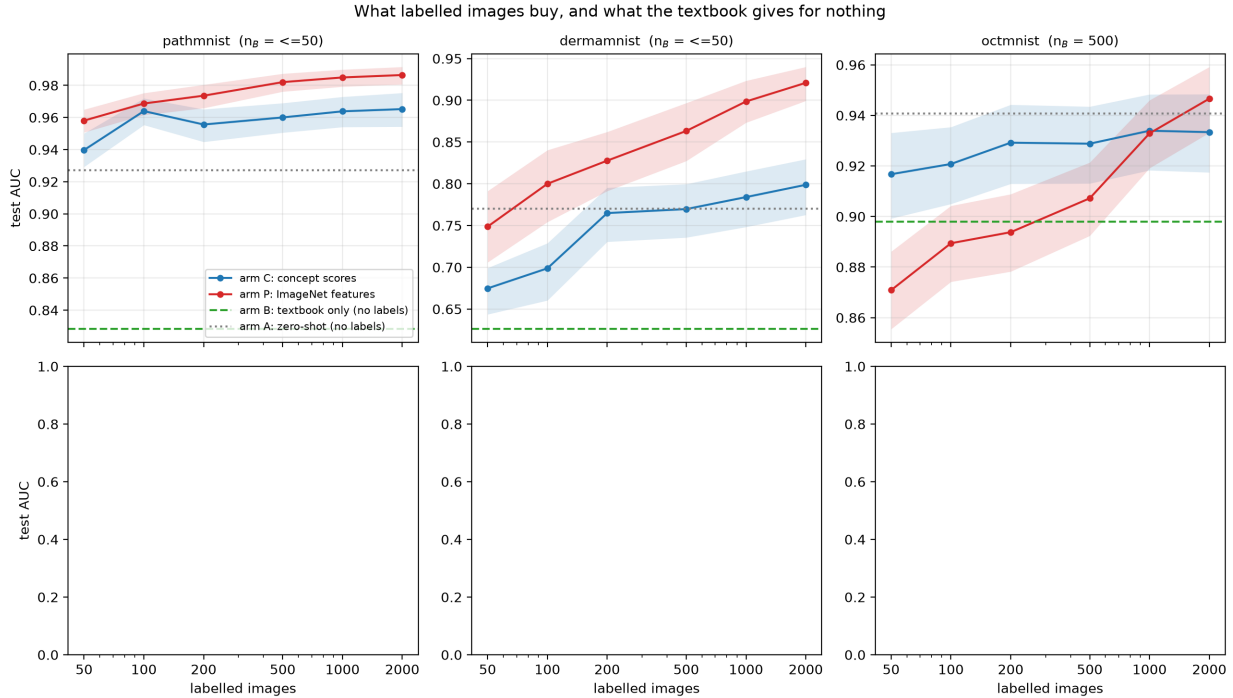


Figure 1: What labelled images buy (arms C and P, with 95% intervals) against what the textbook gives for nothing (arm B, dashed; arm A, dotted). Where the pixel probe crosses arm B’s line is n_B .

dataset	n_B	95% interval	AUC(C, n=50)	AUC(P, n=50)	difference C-P	
pathmnist	≤ 50	$[\leq 50, \leq 50]$	0.940	0.958	-0.018	$[-0.029, -0.007]$
dermamnist	≤ 50	$[\leq 50, \leq 50]$	0.675	0.749	-0.074	$[-0.134, -0.019]$
octmnist	500	$[100, 1000]$	0.917	0.871	0.046	$[0.025, 0.066]$

Table 1: H1. The labelled images the pixel probe needed to reach the textbook arm, and the concept probe against the pixel probe at the smallest labelled subset.

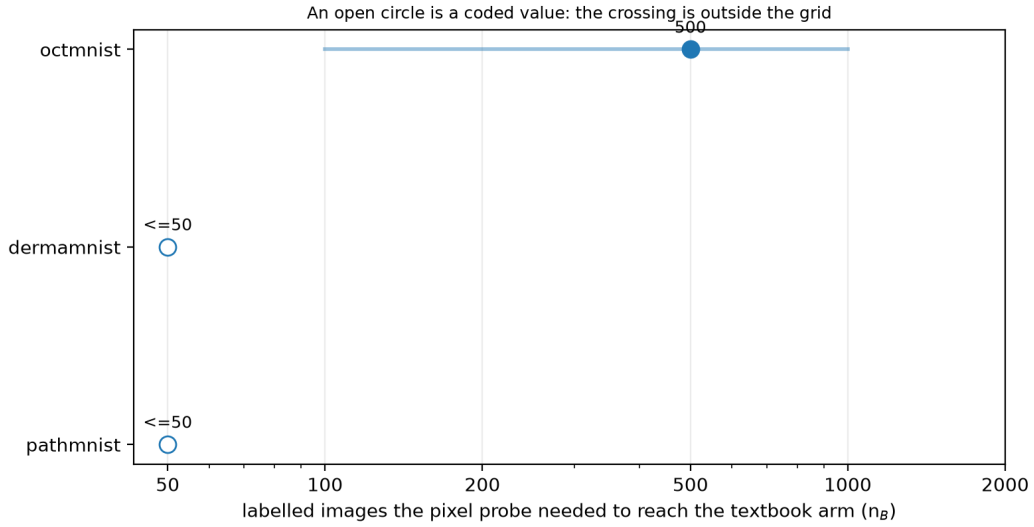


Figure 2: n_B per dataset with its bootstrap interval. An open circle marks a coded value: the crossing lies outside the grid.

H2 — The bank, not just the model: not supported

Directing the model at cited visual features beats asking it for the diagnosis, and the bank’s structure carries the difference. Arm B beat arm A on 0 of 3 datasets ($p = 1.0000$). The two prompts are separate calls on separate prompts — the concept prompt names no class and the zero-shot prompt mentions no concept, which the smoke tier checks on the rendered strings — so the comparison cannot be circular. Table 2 also reports what each arm loses when the structure it claims to read is destroyed.

dataset	AUC(B)	AUC(A)	B – A	B drop, permuted	C drop, permuted
pathmnist	0.924	0.927	-0.003 [-0.019, 0.014]	0.432 [0.415, 0.448]	0.448 [0.432, 0.463]
dermannist	0.671	0.770	-0.099 [-0.168, -0.029]	0.209 [0.153, 0.252]	0.299 [0.255, 0.338]
octmnist	0.894	0.941	-0.046 [-0.064, -0.028]	0.519 [0.495, 0.541]	0.238 [0.219, 0.257]

Table 2: H2. The textbook arm against the zero-shot baseline, and what each arm loses when the structure it claims to read is destroyed: arm B’s fingerprints permuted across classes, arm C’s concept columns permuted across images.

H3 — Scale: not supported

The prior gets better with a bigger model. Read within family, because size and training data are confounded across families and both size steps also change quantisation. The larger model won on 2 of 3 datasets in the qwen family and 2 in the gemma family; a Friedman test over all 4 models gives $p = 0.6149$.

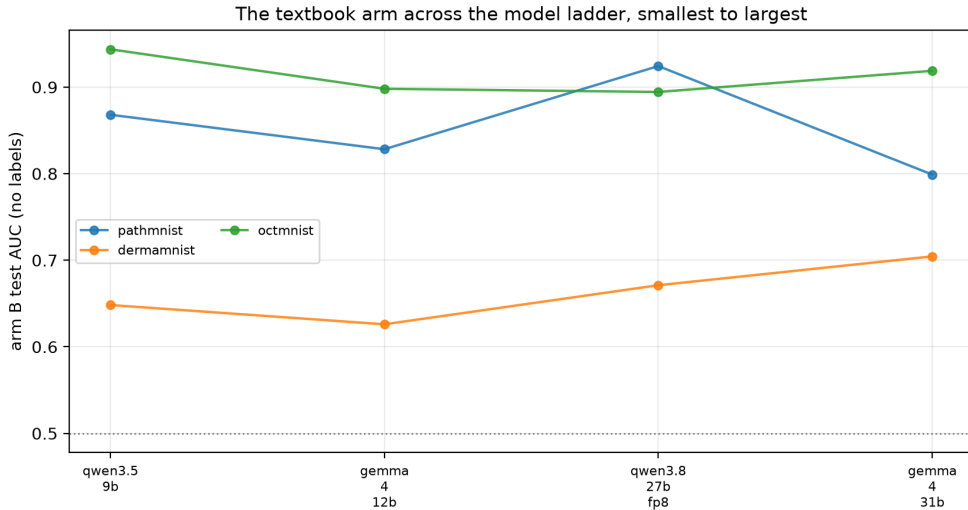


Figure 3: Arm B across the model ladder, smallest to largest.

dataset	qwen3.5-9b	gemma-4-12b	qwen3.8-27b-fp8	gemma-4-31b
pathmnist	0.868	0.828	0.924	0.799
dermamnist	0.648	0.626	0.671	0.704
octmnist	0.944	0.898	0.894	0.919

Table 3: H3. Arm B’s AUC for every model, smallest to largest. Read within family: size and training data are confounded across families, and both size steps also change quantisation.

dataset	gemma-4-12b	gemma-4-31b	qwen3.5-9b	qwen3.8-27b-fp8
pathmnist	100.0	100.0	99.8	100.0
dermamnist	100.0	100.0	100.0	100.0
octmnist	100.0	100.0	100.0	100.0

Table 4: Percentage of test images for which every concept came back with a level on that concept’s own scale. An image with any missing answer counts as incomplete; a cell more than 5% incomplete is excluded from the headline.

5 Diagnostics

6 What this study cannot separate

Every source dataset is public and labelled, so “the model carries textbook knowledge” and “the model has seen this benchmark” are not distinguishable with these data. Not claimed: that the concept scores are clinically valid, that the simulated review substitutes for a clinician, or that any arm is state of the art.

7 Run record

Every result file carries a manifest: its parameters, its seeds, the git commit and dirty flag, package versions, host, thread pins and wall time. The response archive additionally records the served model name, the prompt hash and the bank hash for every call, and is write-protected once written, so re-querying is a deliberate act.

Sources behind the concept bank

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